The examination is being carried out on the **following application documents**

Description, Pages

1-35 filed with entry into the regional phase before the EPO

Claims, Numbers

1-13 filed with entry into the regional phase before the EPO

Drawings, Sheets

1/9-9/9 filed with entry into the regional phase before the EPO

1 Reference is made to the following documents; the numbering will be adhered to in the rest of the procedure.

D1 CN 210 142 251 U (NINGBO WISSEN INTELLIGENT SENSING TECH CO LTD) 13 March 2020 (2020-03-13)

D2 JP 2010 221572 A (PANASONIC ELEC WORKS CO LTD) 7 October 2010 (2010-10-07)

D3 JP 2019 155694 A (NIDEC COPAL CORP) 19 September 2019 (2019-09-19)

independent claim 1

2 D1 discloses (cf. the wording of claim 1) a sensor module, comprising:

a) a sensor element (24) [fig. 1: sensor 301];

b) a first case (11) [fig. 1: first casing 1] that includes an opening end including a first welded region and accommodates therein the sensor element;

c) a second case (12) [fig. 1: second casing 2] that includes a joining surface including a second welded region, the second welded region being welded to the first welded region to form a welded portion; and

d) a groove (70) [fig. 1: receiving groove 2021] that is formed in at least one of the first welded region or the second welded region.

2.1 Therefore, D1 discloses all the features defined in claim 1, such that this claim lacks novelty.

dependent claims

3 The dependent claims 2 - 4 and 9 - 13 define features which are either known from D1, or are well known to a person skilled in the art and may be applied without inventive thinking to solve different well-known problems. In particular, reference is made to the comments concerning the claims below.

3.1 claims 2 and 3: see D1, fig. 3

3.2 claim 4: This claim further defines the feature *wherein the groove includes a plurality of first grooves formed in parallel over the entire circumference of the first welded region or the second welded region*.

3.2.1 The objective problem may be defined as how to improve a laser welding joint between two resins such as to avoid resin material from protruding and achieving a more stable welding strength.

3.2.2 D2 discloses the idea of increasing the number of grooves (D2, Fig. 6: 501a) in order to avoid resin material from protruding and achieving a more stable welding strength.

3.2.3 A person skilled in the art having knowledge of D1 and D2 would consider adapting D1 by increasing the number of grooves.

3.2.4 Therefore, claim 4 is considered to lack an inventive step.

3.3 claim 9: see D3, Fig. 7: two surfaces facing each other and having a gap in-between for providing a space for protruding molten material

3.4 claim 10: see D1:

a) wherein the first case is made of a resin material that absorbs laser [D1: "*The first housing 1 and the lens are an integrated structure, and a laser-absorbing plastic is used, so that the laser does not pass through the first housing 1 and prevents the laser from penetrating into the module and affecting the imaging quality*"] light of the predetermined wavelength, and

b) the second case is made of a resin material that transmits the laser light of the predetermined wavelength. [D1: *"the second casing 2 includes a second base body 201 and a ring-shaped mounting portion 202, and the material is a laser-transmitting plastic."*]

3.5 claim 11: see D1: Fig. 8 shows molten resin material.

3.6 claim 12: see D1, title: "camera module"

3.7 claim 13: see D1: *"the first casing 1 and the second casing 2 are fixed by laser welding"*

Inventive subject-matter

4 Claims 5 - 8 define different features concerning how to arrange grooves such as to avoid superfluous laser welding material from protruding out of the laser welding joint.

5 Therefore, claim 5 - 8 are considered to involve an inventive step.

Further comments

6 The features of the claims should be provided with reference signs placed in parentheses to increase the intelligibility of the claims (Rule 43(7) EPC). This applies to both the preamble and characterising portion (see EPC Guidelines F-IV, 4.18).

7 To meet the requirements of Rule 42(1)(b) EPC, document D1 should be identified in the description and the relevant background art disclosed therein should be briefly discussed.